

CLASS 9 MATHEMATICS

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CHAPTER 5 – CIRCLES

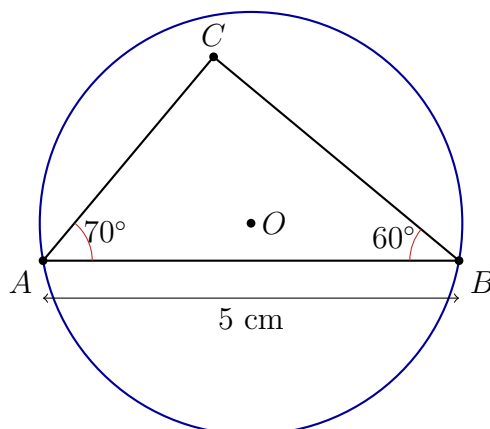
EXERCISE 5.1

Question 1

Draw $\triangle ABC$ with $AB = 5$ cm, $\angle A = 70^\circ$ and $\angle B = 60^\circ$. Draw the circumcircle of $\triangle ABC$. Is the centre inside or outside the triangle?

Construction

1. Draw $AB = 5$ cm.
2. At A , construct $\angle A = 70^\circ$.
3. At B , construct $\angle B = 60^\circ$.
4. Let the two rays meet at C . Thus, $\triangle ABC$ is obtained.
5. Draw the perpendicular bisectors of AB and AC .
6. Let them intersect at O .
7. With O as centre and OA as radius, draw the circumcircle.



Solution

We have

$$\angle A = 70^\circ, \quad \angle B = 60^\circ.$$

Therefore,

$$\angle C = 180^\circ - (70^\circ + 60^\circ) = 50^\circ.$$

Thus,

$$\angle A < 90^\circ, \quad \angle B < 90^\circ, \quad \angle C < 90^\circ.$$

Hence, $\triangle ABC$ is an acute-angled triangle.

The circumcentre of an acute-angled triangle lies **inside** the triangle.

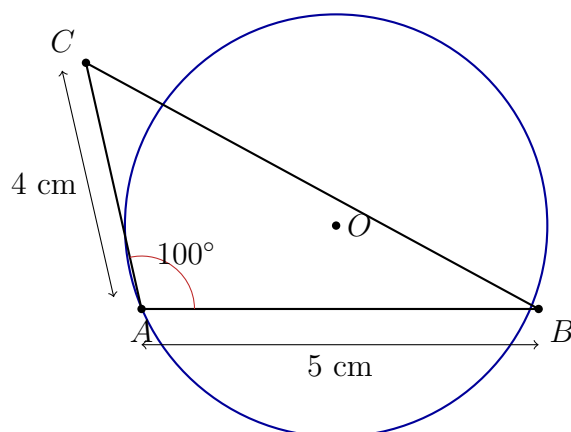
Answer: The circumcentre O lies inside the triangle.

Question 2

Draw $\triangle ABC$ with $AB = 5$ cm, $\angle A = 100^\circ$, $AC = 4$ cm. Draw the circumcircle of $\triangle ABC$. Is the centre inside or outside the triangle?

Construction

1. Draw $AB = 5$ cm.
2. At A , construct an angle of 100° .
3. On this ray, mark $AC = 4$ cm.
4. Join C to B .
5. Draw the perpendicular bisectors of two sides of the triangle.
6. Let them meet at O .
7. With O as centre and OA as radius, draw the circumcircle.



Solution

Given,

$$\angle A = 100^\circ.$$

Since

$$100^\circ > 90^\circ,$$

the triangle is an obtuse-angled triangle.

The circumcentre of an obtuse-angled triangle lies **outside** the triangle.

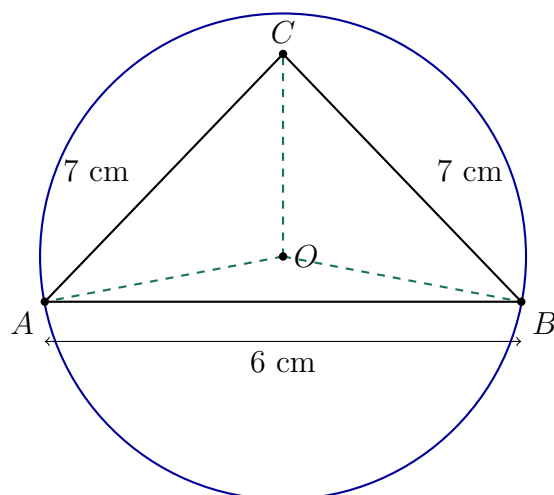
Answer: The circumcentre O lies outside the triangle.

Question 3

Draw $\triangle ABC$, with $AB = 6$ cm, $BC = 7$ cm and $CA = 7$ cm. Draw the circumcircle of $\triangle ABC$. Let the circumcentre be O . Measure OA , OB , OC .

Construction

1. Draw $AB = 6$ cm.
2. With A as centre and radius 7 cm, draw an arc.
3. With B as centre and radius 7 cm, draw another arc.
4. Let the arcs meet at C .
5. Join AC and BC .
6. Draw perpendicular bisectors of AB and AC .
7. Let them intersect at O .
8. With O as centre and OA as radius, draw the circumcircle.



Solution

Since O is the circumcentre, it is equidistant from all three vertices of the triangle.
Therefore,

$$OA = OB = OC.$$

For the given triangle,

$$AB = 6 \text{ cm}, \quad AC = BC = 7 \text{ cm}.$$

The circumradius is approximately

$$R = OA = OB = OC \approx 4.33 \text{ cm}.$$

Hence, by measurement,

$$OA \approx 4.33 \text{ cm},$$

$$OB \approx 4.33 \text{ cm},$$

$$OC \approx 4.33 \text{ cm}.$$

Answer:

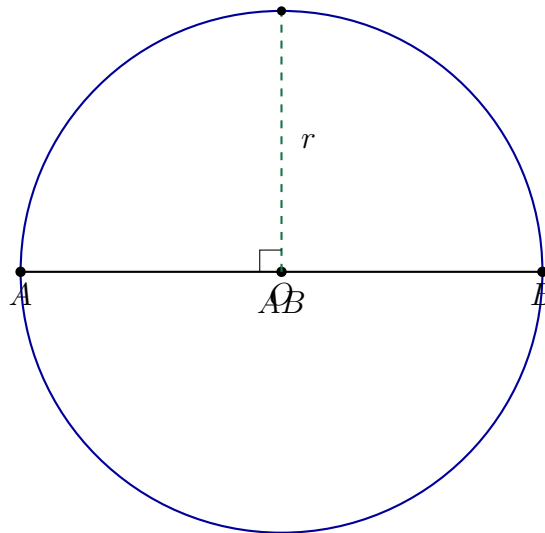
$$OA \approx OB \approx OC \approx 4.33 \text{ cm}$$

The three measurements are equal because O is the circumcentre.

Question 4

What is the least possible radius of a circle through two points A and B ?

Diagram



Solution

Let the radius of the circle be r .

Since the circle passes through both A and B ,

$$OA = OB = r.$$

The distance between two points A and B cannot be greater than the diameter of the circle. Therefore,

$$AB \leq 2r.$$

Hence,

$$r \geq \frac{AB}{2}.$$

The least value of r occurs when AB is the diameter of the circle.

Thus,

$$\text{Least possible radius} = \frac{AB}{2}$$

This happens when the centre O is the midpoint of AB .

QUICK REVISION

Position of Circumcentre

- Acute-angled triangle:

Circumcentre lies inside

- Right-angled triangle:

Circumcentre is the midpoint of hypotenuse

- Obtuse-angled triangle:

Circumcentre lies outside

Also,

$$OA = OB = OC$$

because all three points A, B, C lie on the same circumcircle.

For a circle passing through two points A and B ,

$$r_{\min} = \frac{AB}{2}.$$